

Q.1 FULL INNER JOIN.

Query:

```
SELECT  
  
S1.SUB_ID,  
  
S1.SUB_NAME,  
  
C1.ROOM_NAME  
  
FROM SUBJECT S1  
  
LEFT JOIN CLASSROOM C1 ON S1.SUB_ID=C1.SUB_ID  
  
UNION  
  
SELECT C1.CLASS_ID,  
  
C1.ROOM_NAME,  
  
S1.SUB_ID  
  
FROM CLASSROOM C1  
  
RIGHT JOIN SUBJECT S1 ON  
  
C1.SUB_ID=S1.SUB_ID;
```

Screenshot:

The screenshot displays the MySQL Workbench interface. The main window shows a SQL query being executed. The query is as follows:

```
SELECT  
S1.SUB_ID,  
S1.SUB_NAME,  
C1.ROOM_NAME  
FROM SUBJECT S1  
LEFT JOIN CLASSROOM C1 ON S1.SUB_ID=C1.SUB_ID  
UNION  
SELECT C1.CLASS_ID,  
C1.ROOM_NAME,  
S1.SUB_ID  
FROM CLASSROOM C1  
RIGHT JOIN SUBJECT S1 ON  
C1.SUB_ID=S1.SUB_ID;
```

The query is executed, and the results are displayed in the 'Result Grid' tab. The results show a list of subjects and their corresponding room names, with the first column being the subject ID and the second column being the room name.

SUB_ID	SUB_NAME	ROOM_NAME
CS101	Introduction to Database	101-A
CS102	Artificial Intelligence	102-B
HI101	Ancient Civilizations	103-C
MA301	Linear Algebra	201
PH301	Quantum Mechanics	Lab-3

The 'Output' tab shows the execution details, including the time taken for each statement and the number of rows returned. The first statement (SELECT S1.SUB_ID, S1.SUB_NAME, C1.ROOM_NAME FROM SUBJECT S1 LEFT JOIN CLASSROOM C1 ON S1.SUB_ID=C1.SUB_ID) returned 10 rows. The second statement (SELECT C1.CLASS_ID, C1.ROOM_NAME, S1.SUB_ID FROM CLASSROOM C1 RIGHT JOIN SUBJECT S1 ON C1.SUB_ID=S1.SUB_ID) returned 10 rows.

Insights:

- Room name for HI101 is empty since the classroom table doesn't contain the HI101 as a value in SUB_ID.
- Though SUB_ID HI101 is fetched, as per the union operator's function.
- As it fetches the unique data removing duplications and combines the rows of both the tables.